

MODULE 3

ADDRESSING

In addition to the physical address the internet requires an additional addressing convention: an address that identifies the connection of a host to its network.

Each Internet address consists of 4 bytes defining three fields: class type, netid, and hostid. These parts are varying lengths depending on the class of the address.

CLASSES

There are currently five different classes:

They are Class A, Class B, Class C, Class D, Class E

Class A: This can accommodate more hosts since 3 bytes are reserved for HOSTID. Class A will begin with 0 .

Class B :

This will start with **10** and Host id will have 2 bytes length.

Class C :

This will start with **110** and Hostid will have 1 byte length.

Class D:

This will start with **1110** . This is reserved for **Multicast addresses**.

Class E :

This is reserved for future use and will start with **1111** .

Ex. for classes

1. 01111011 10001111 11111100 11001111



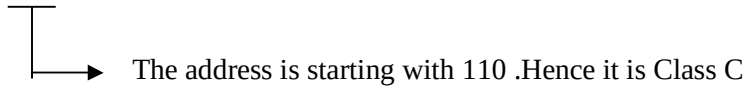
The address is starting with 0 .Hence it is Class A.

2. 10011101 10001111 11111100 11001111



The address is starting with 10 .Hence it is Class B.

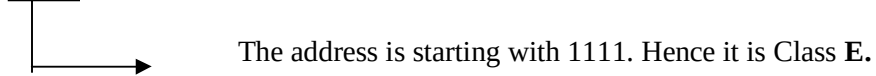
3. 11011101 10001111 11111100 11001111



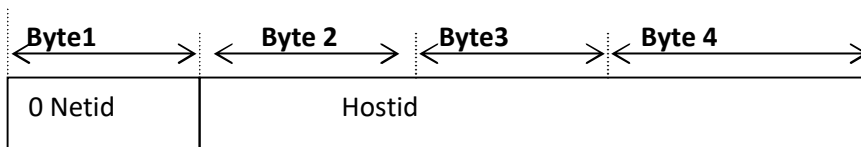
4. 11101011 10001111 11111100 11001111



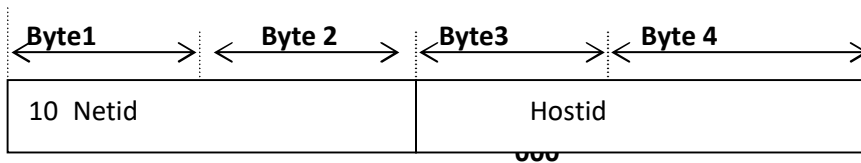
5. 11110101 10001111 11111100 11001111



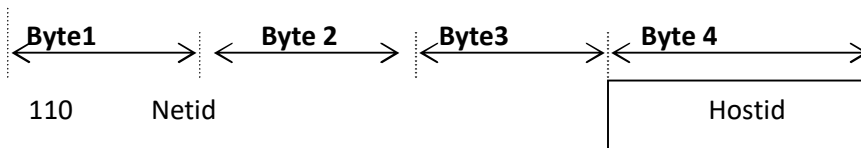
CLASS A :



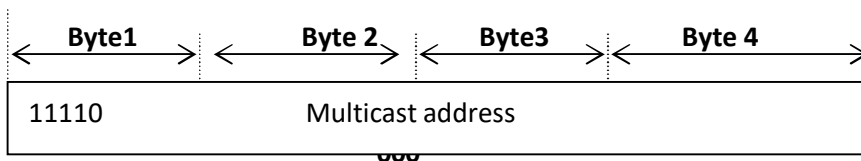
CLASS B:



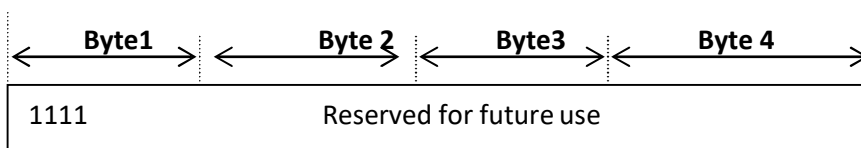
Class C :



Class D :

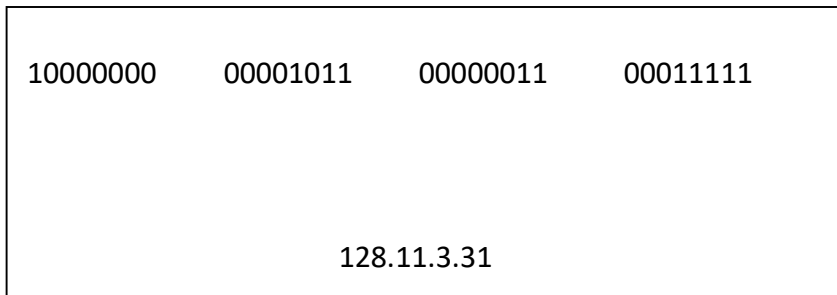


Class E :



Dotted-Decimal Notation:

To make 32 bit form shorter and easier to read, Internet addresses are usually written in decimal form with decimal points separating the bytes – dotted – decimal notation.



IP addresses in decimal notation

Class ranges of Internet address

From	To	
0.0.0.0	127.255.255.255	← Class A
From	To	
128.0.0.0	191.255.255.255	← Class B
From	To	
192.0.0.0	223.255.255.255	← Class C
From	To	
224.0.0.0	239.255.255.255	← Class D
From	To	
240.0.0.0	255.255.255.255	← Class E

Goals of IPv6

- Huge address space
- Reduce the size of routing tables (possibly at the cost of wasting address space)
- Simplify protocol to speed routers
- Better security
- Better type-of-service (to support realtime)
- Aid multicasting
- Allow roaming hosts to maintain a constant address
- Allow for future evolution

Version (4-bits) : Indicates version of Internet Protocol which contains bit sequence 0110.

Traffic Class (8-bits) : The Traffic Class field indicates class or priority of IPv6 packet which is similar to *Service Field* in IPv4 packet. It helps routers to handle the traffic based on priority of the packet. If congestion occurs on router then packets with least priority will be discarded.

As of now only 4-bits are being used (and remaining bits are under research), in which 0 to 7 are assigned to Congestion controlled traffic and 8 to 15 are assigned to Uncontrolled traffic.

Priority assignment of Congestion controlled traffic :

Priority	Meaning
0	No Specific traffic
1	Background data
2	Unattended data traffic
3	Reserved
4	Attended bulk data traffic
5	Reserved
6	Interactive traffic
7	Control traffic

Uncontrolled data traffic is mainly used for Audio/Video data. So we give higher priority to Uncontrolled data traffic. Source node is allowed to set the priorities but on the way routers can change it. Therefore, destination should not expect same priority which was set by source node.

Flow Label (20-bits) : Flow Label field is used by source to label the packets belonging to the same flow in order to request special handling by intermediate IPv6 routers, such as non-default quality of service or real time service. In order to distinguish the flow, intermediate router can use source address, destination address and flow label of the packets. Between a source and destination multiple flows may exist because many processes might be running at the same time. Routers or Host that do not support the functionality of flow

label field and for default router handling, flow label field is set to 0. While setting up the flow label, source is also supposed to specify the lifetime of flow.

Payload Length (16-bits) : It is a 16-bit (unsigned integer) field, indicates total size of the payload which tells routers about amount of information a particular packet contains in its payload. Payload Length field includes extension headers(if any) and upper layer packet. In case length of payload is greater than 65,535 bytes (payload up to 65,535 bytes can be indicated with 16-bits), then the payload length field will be set to 0 and jumbo payload option is used in the Hop-by-Hop options extension header.

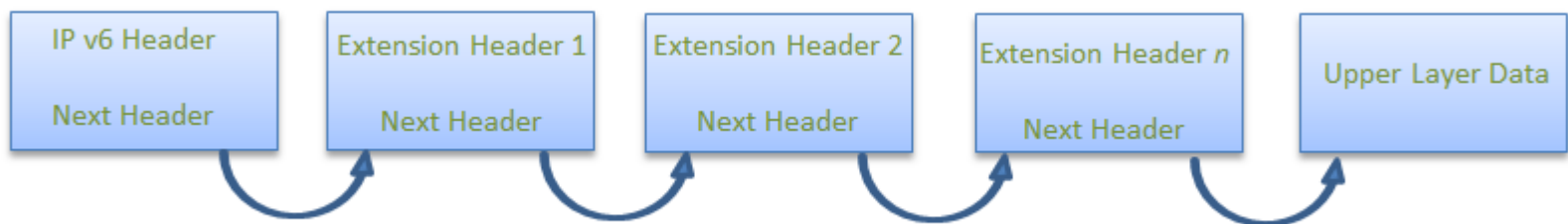
Next Header (8-bits) : Next Header indicates type of extension header(if present) immediately following the IPv6 header. Whereas in some cases it indicates the protocols contained within upper-layer packet, such as TCP, UDP.

Hop Limit (8-bits) : Hop Limit field is same as TTL in IPv4 packets. It indicates the maximum number of intermediate nodes IPv6 packet is allowed to travel. Its value gets decremented by one, by each node that forwards the packet and packet is discarded if value decrements to 0. This is used to discard the packets that are stuck in infinite loop because of some routing error.

Source Address (128-bits) : Source Address is 128-bit IPv6 address of the original source of the packet.

Destination Address (128-bits) : Destination Address field indicates the IPv6 address of the final destination(in most cases). All the intermediate nodes can use this information in order to correctly route the packet.

Extension Headers : In order to rectify the limitations of *IPv4 Option Field*, Extension Headers are introduced in IPversion 6. The extension header mechanism is very important part of the IPv6 architecture. Next Header field of IPv6 fixed header points to the first Extension Header and this first extension header points to the second extension header and so on.



IPv6 packet may contain zero, one or more extension headers but these should be present in their recommended order:

Order	Header Type	Next Header Code
1	Basic IPv6 Header	-
2	Hop-by-Hop Options	0
3	Destination Options (with Routing Options)	60
4	Routing Header	43
5	Fragment Header	44
6	Authentication Header	51
7	Encapsulation Security Payload Header	50
8	Destination Options	60
9	Mobility Header	135
	No next header	59
Upper Layer	TCP	6
Upper Layer	UDP	17
Upper Layer	ICMPv6	58

Example: TCP is used in IPv6 packet

Next Header= 6	TCP header	TCP data
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Example2:

Next Header= 43	Routing Extension Header Next Header= 6	TCP header	TCP data
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Rule : Hop-by-Hop option header(if present) should always be placed after IPv6 base header.

Conventions :

- Any extension header can appear at most once except Destination Header because Destination Header is present two times in above list itself.
- If Destination Header is present before Routing Header then it will be examined by all intermediate nodes specified in routing header.
- If Destination Header is present just above Upper layer then it will be examined only by Destination node.

Given order in which all extension header should be chained in IPv6 packet and working of each extension header :

Ext. Header	Description
Hop-by-Hop Options	Examined by all devices on the path
Destination Options (with routing options)	Examined by destination of the packet
Routing Header	Methods to take routing decision
Fragment Header	Contains parameters of fragmented datagram done by source
Authentication Header	verify authenticity
Encapsulating Security Payload	Carries Encrypted data

Example:

Find the class of each address a.

- 4.23.145.90
- b. 227.34.78.7
- c. 246.7.3.8
- d. 129.6.8.4
- e. 198.76.9.23

The first byte defines the class.

- a. The binary equivalent for 4 is 0000 0100. Since the first bit is 0 ,it is CLASS A.
- b. The binary equivalent for 227 is 11100011 . Since it is starting with 111 ,it is CLASS D.
- c. The binary equivalent for 246 is 11110110 . Since it is starting with 111 ,it is CLASS E.
- d. The binary equivalent for 129 is 10000001 . Since it is starting with 10,it is CLASS B.
- e. The binary equivalent for 198 is 11000110 . Since it is starting with 110 , it is CLASS C.

Example:

Find the netid and hosted for each address : a.

- 4.23.145.90
- b. 227.34.78.7
- c. 246.7.3.8
- d. 129.6.8.4
- e. 198.76.9.23

First find the class and then netid and hosted.

- a. Class A , netid :4 , hosted : 23.145.90
- b. Class D , no netid or hosted;
- c. Class E , no netid , or hosted;
- d. Class B , netid :129.6 ,hosted :8.4 ;
- e. Class C netid : 198.76.9 hostid 23 ;

TCP Protocol

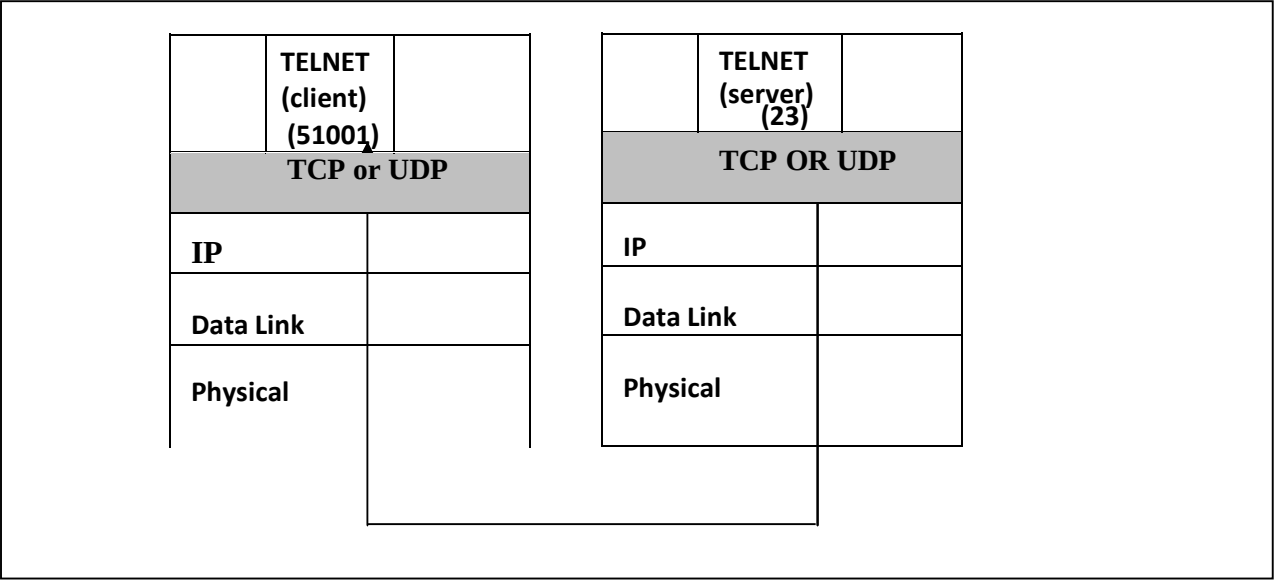
ARPA established a packet-switching network of computers linked by point-to-point leased lines called Advanced Research Project Agency Network(ARPANET) that provided a basics for early research into networking. The conventions developed by ARPA to specify how individual computers could communicate across that network became TCP/IP.

The transport layer is represented in TCP/IP by two protocols:TCP and UDP. Of these, UDP is similar; it provides nonsequenced transport functionality when reliability and security are less important than size and speed.

The transport protocols of the TCP/IP suite define a set of conceptual connections to individual process called protocol ports or, more simply, ports. A protocol port is a destination point (usually a buffer) for storing data for use by a particular process.

The IP is a host-to-host protocol, meaning that it can deliver a packet from one physical device to another. TCP/IP’s transport level protocols are port-to-port protocols that work on top of the IP protocols to deliver the packet from the originating port to the IP services at the start of a transmission, and from the IP services to the destination port at the start end.

Figure Port addresses



Each port is defined by a positive integer address carried in the header of a transport layer packet. An IP datagram uses the host’s 32-bit Internet address. A frame at the transport level uses the process port address of 16 bits, enough to allow the support of up to 65,536(0 to 65535) ports.

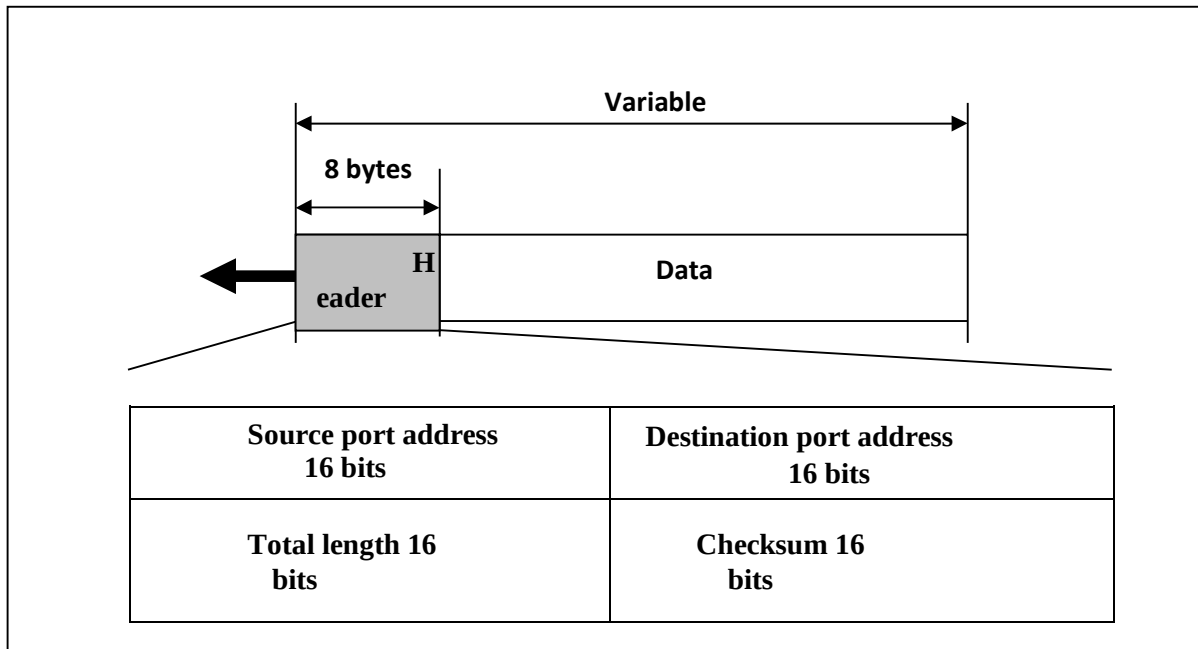
USER DATAGRM PROTOCOL(UDP)

The user datagram protocol (UDP) is the simpler of the two standard TCP/IP transport protocols. It is an end-to-end transport level protocol that adds only port addresses, check sum error control, and length information

to the data from the upper layer. The packet produced by the UDP is called a user datagram .

- **Source port address.** The source port address is the address of the application program that has created the message.
- **Destination port address.** The destination port address is the address of the application program that will receive the message.
- **Total length.** The total length field defines the total length of the user datagram in bytes.
- **Check sum.** The check sum is a 16-bit field used in error detection.

Figure UDP datagram format



UDP provides only the basic functions needed for end-to-end delivery of a transmission. It does not provide any sequencing or recording functions and cannot specify the damaged packet when reporting an error (for which it must be paired with ICMP). UDP can discover that an error has occurred; ICMP can then inform the sender that a user datagram has been damaged and discarded. Neither, however, has the ability to specify which packet has been lost. UDP contains only a checksum; it does not contain an ID or sequencing number for a particular data segment.

Transmission Control Protocol(TCP)

The Transmission Control Protocol (TCP) provides full transport layer services to applications. TCP is a reliable stream transport port-to-port protocol. The term stream, in this context, means connection-oriented: a connection must be established between both ends of a transmission before either may transmit data. By creating this connection, TCP generates a virtual circuit between sender and receiver that is active for the duration of a transmission.(connections for the duration of an entire exchange are different, and are handled

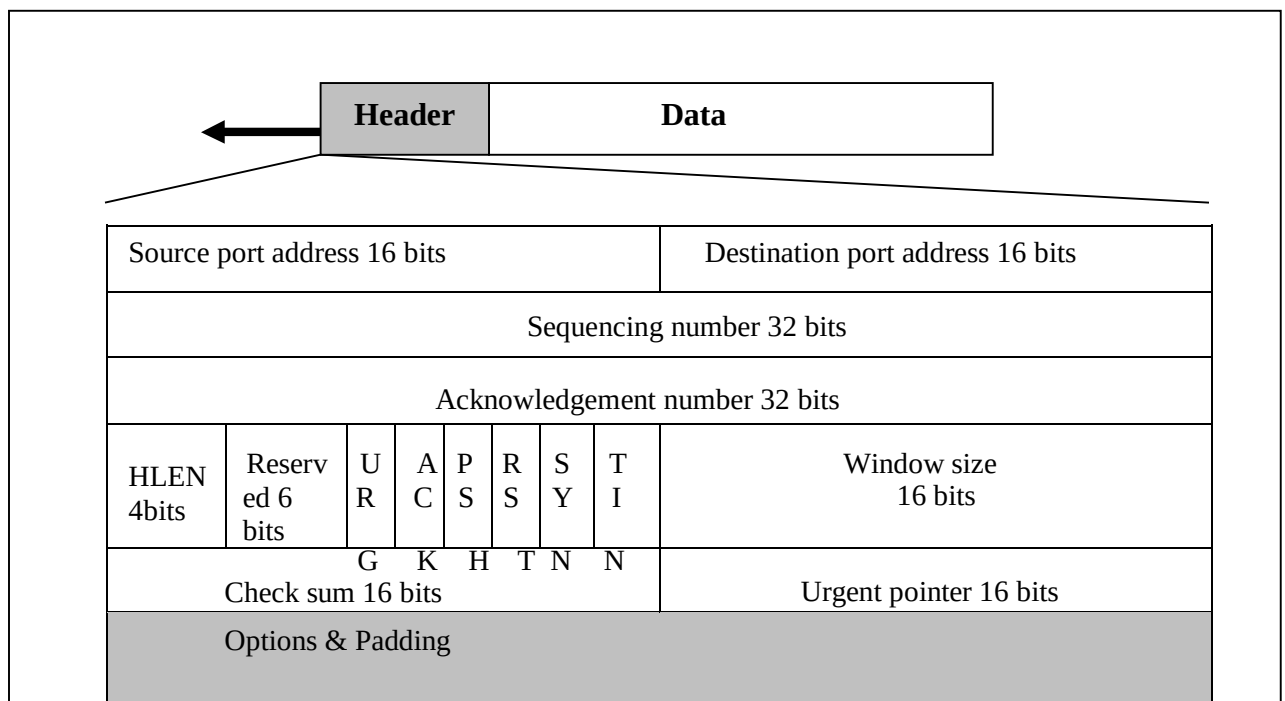
by session functions in individual applications.) TCP begins each transmission by altering the receiver that datagrams are on their way (connection establishment) and ends each transmission with a connection termination. In this way, the receiver knows to expect the entire transmission rather than a single packet.

IP and UDP treat multiple datagrams belonging to a single transmission as entirely separate units, unrelated to each other. The arrival of each datagram at the destination is therefore a separate event, unexpected by the receiver. TCP, on the other hand, as a connection-oriented service, is responsible for the reliable delivery of the entire stream of bits contained in the message originally generated by the sending application. Reliability is ensured by provision for error detection and retransmission of damaged frames; all segments must be received and acknowledged before the transmission is considered complete and the virtual circuit is discarded. At the sending end of each transmission, TCP divides long transmissions into smaller data units and packages each into a frame called a segment. Each segment includes a sequencing number for reordering after receipt, together with an acknowledgement ID number and a window-size field for sliding window ARQ. Segments are carried across network links inside of IP datagrams as it comes in and reorders the transmission based on sequence numbers.

The TCP Segment

The scope of the services provided by TCP requires that the segment header be extensive. A comparison of the TCP segment format with that of a UDP user datagram shows the differences between the two protocols. TCP provides a comprehensive range of reliability functions but sacrifices speed (connections must be established, acknowledgments waited for , etc.).Because of its smaller frame size, UDP is much faster than TCP, but at the expense of reliability. A brief description of each field is in order.

Figure TCP Segment format



- **Source port address.** The source port address defines the application program in the source computer.
- **Destination port address.** The destination port address defines the application program in the destination computer.
- **Sequence number.** A stream of data from the application program may be divided into two or more TCP segments. The sequence number field shows the position of the data in the original data stream.
- **Acknowledgement number.** The 32-bit acknowledgement number is used to acknowledge the receipt of data from the other communicating device. This number is valid only if the ACK bit in the control field(explained later) is set. In this case, it defines the byte sequence number that is next expected.
- **Header Length (HLEN).** The four-bit HLEN field indicates the number of 32-bit (four-byte) words in the TCP header. The four bits can define a number up to 15.This is multiplied by 4 to give the total number of bytes in the header. Therefore, the size of the header can be a maximum of 60 bytes (4x15).Since the minimum required size of the header is 20 bytes, 40 bytes are thus available for the options section.
- **Reserved.** A six-bit field is reserved for future use.
- **Control.** Each bit of the six-bit control field functions individually and independently. A bit can either define the use of a segment or serve as a validity check for other fields. The urgent bit, when set, validates the urgent pointer field. Both this bit and the pointer indicate that the data in the segment are urgent. The ACK bit, when set, validates the acknowledgement number field. Both are used together and have different functions, depending on the segment type. The PSH bit is used to inform the sender that a higher throughput is needed. If possible, data must be pushed through paths with higher throughput. The reset bit is used to reset the connection when there is confusion in the sequence numbers. The SYN bit is used for sequence number synchronization in three types of segments: connection request, connection confirmation (with the ACK bit set), and confirmation acknowledgement (with the ACK bit set). The FIN bit is used in connection termination in three types of segments: termination request, termination confirmation (with the ACK bit set), and acknowledgement of termination confirmation (with the ACK bit set).
- **Window size.** The window is a 16-bit field that defines the size of the sliding window.
- **Checksum.** The checksum is a 16-bit field used in error detection.
- **Urgent pointer.** This is the last required field in the header. Its value is valid only if the URG bit in the control field is set. In this case, the sender is informing the receiver that there are **urgent data** in the data portion of the segment. This pointer defines the end of urgent data and the start of normal data.
- **Options and padding.** The remainder of the TCP header defines the optional fields. They are used

to convey additional information to the receiver or for alignment purposes.

Functions of Net Work layer

1. Routing
2. Congestion Control

Routing algorithms

The main function of the network layer is routing packets from the source machine to the destination machine. Routing algorithm can be grouped into two major classes. Nonadaptive and Adaptive algorithms.

Non adaptive

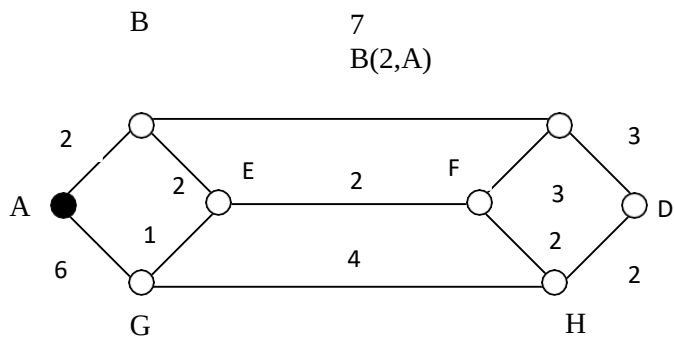
- 1) Routing decisions are not based on measurements or estimates of the current traffic and topology.
- 2) The route is computed well in advance.
- 3) When the network is booted the routers are downloaded.
- 4) This is a static routing.

Adaptive

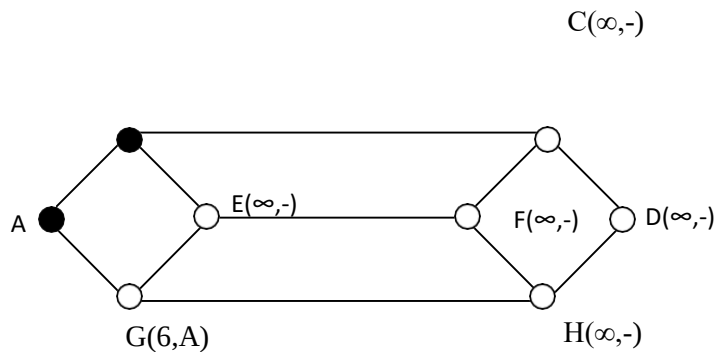
- 1) Routing decisions are based on measurements of the current traffic and topology.
- 2) The route is computed depends on situation.
- 3) The routers are not downloaded.
- 4) This is a dynamic routing.

Shortest Path Routing:

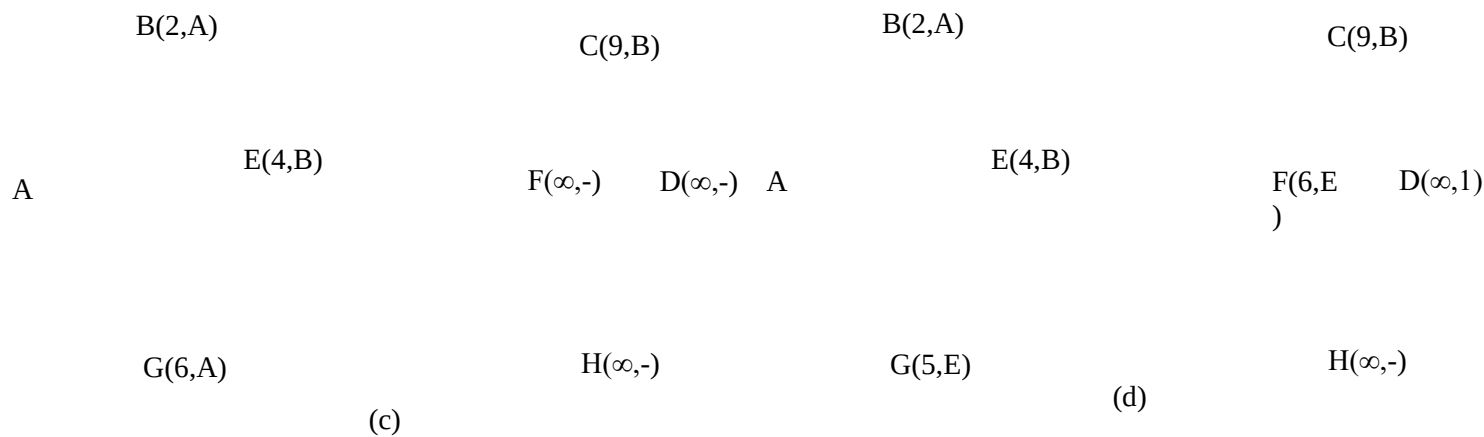
This is a static routing algorithm. The idea is to build a graph of the subnet, with each node of the graph representing a router and each arc of the graph representing a communication line. To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph.



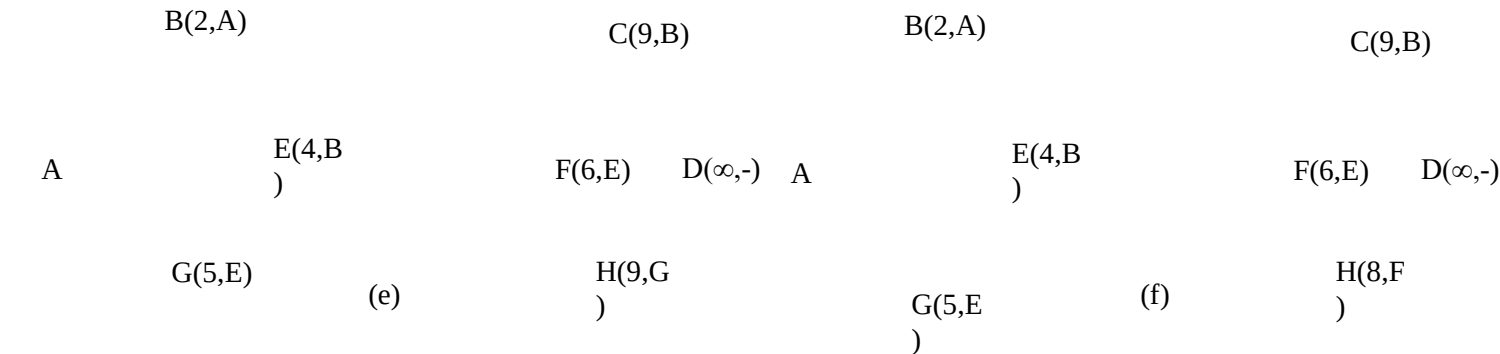
(a)



(b)



(c)



(d)



(e)



(f)

One-way of measuring path length is the number of hops. Using this metric, the paths ABC and ABE are equally long. (Two hops).

Another metric is the Geographic distance in Kilometers. ABC is clearly longer than ABE.

Many other metrics are also possible besides hops and physical distance. Each are could be labeled with the mean queuing and transmission delay for some standard test packets as determined by hourly test runs. With this graph labeling, the shortest path is the fastest path, rather than the path with the fewest arc or kilometers.

In most general case, the labels on the arcs could be computed as a function of the distance, bandwidth, average traffic, communication cost, mean queue length, measured delay and other factors.

The shortest path can be calculated using Dijkstra method. Each node is labeled with its distance from the source along the best known path. Initially, no paths are known, so all nodes are labeled with infinity. As the algorithm proceeds and paths are found, the labels may change, reflecting better paths. Initially all labels are tentative. When it is discovered that a label represents the shortest path from the source to that node, it is made permanent and never changed thereafter.

In the above diagram, let the weights represents the distance. To find out the shortest path from A to D. We start by marking A as permanent. The examine each one with the distance to A, relabeling each one with the distance to A. Whenever a node is relabeling also label it with the node from which the probe was made. After examing each of the nodes adjacent to A, examine all the tentatively labeled nodes in the whole graph and make the one with the smallest label permanent. This one becomes the new working node.

The same procedure is adopted to all the nodes and the shortest path is found.

Flooding:

This is a static algorithm. In this, every incoming packet is sent out on every outgoing line except the one it arrived on. Flooding will generate vast numbers of duplicate packets, some measures have to take to dump the duplicate packets. One such measure is to have a hop counter contained in the header of each packet, which is decremented at each hop, with the packet being discarded when the counter reaches zero. The hop counter should be initialized to the length of the path from source to destination. If the sender does not know how long the path is it can initialize the counter to full diameter of the subnet.

A variation of flooding is 'Selective Flooding'. In this the routers do not send every incoming packet on every line, instead only on those lines that are going approximately in the right direction which leads to the destination.

Advantages

- 1) In military applications, where large numbers of routers are blown, flooding is desirable.
- 2) In Distributed database applications, it is some times necessary to update all the databases concurrently, in which flooding is useful.
- 3) It is used as a metric against which other routing algorithms are compared.
- 4) Flooding chooses the shortest path, because it chooses all possible path in parallel.

Flow-based Routing:

The flooding and shortest path algorithm takes the topology in to account. Flow based routing algorithm uses both topology and load for routing.

In some networks the mean data flow between each pair of nodes is relatively stable and predictable. The average traffic is known in advance and to a reasonable approximation, constant in time, it is possible to analyze the flows mathematically to optimize routing.

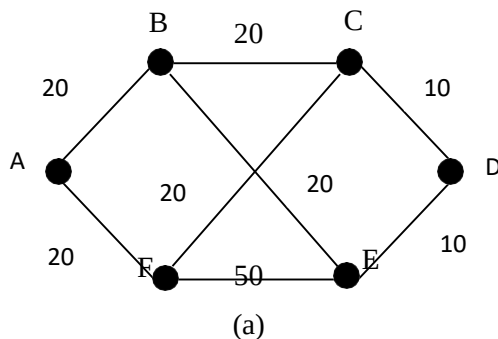
The basic idea behind this, for a given line, if the capacity and average flow are known, it is possible to compute the mean packet delay on that line. From the mean delays on all the lines, the mean packet delay for the whole subnet.

To use this technique, certain information must be known in advance. 1) About the subnet topology 2) about the traffic 3) the line capacity 4) a routing algorithm

The fig.(b) gives the information packets /sec go from source i to destination j. Given this information, it is easy to calculate the total in line for i i.e. λ_i . Using the traffic from the source to destination the mean number of packets/sec on each line. λ_{ci} can be calculated, assuming a mean packet size $1/\mu$. The mean delay for each line can be derived where $1/\mu$ is the mean packet size in bits, μ is the mean flow in packets/sec.

With a capacity $\mu_c = 25$ packets/sec and an actual flow $\mu = 14$ packets/sec, the mean delay is 91 m sec. When $\mu = 0$, the mean delay is 40m sec. With this example we can say the delay depends on both queuing and service time.

The mean delay time for the entire subnet can be calculated as the sum of each of the eight lines, with the weight being the fraction of the total traffic using that line.



	Destination					
	A	B	C	D	E	F
A		9 AB	4 ABC	1 ABFD	7 AE	4 AEF
B	9 AB		8 BC	3 BFD	2 BFE	4 BF
C	4 CBA	8 CB		3 CD	3 CE	2 CEF
D	1 DFBA	3 DFB	3 DC		3 DCE	4 DF
E	7 EA	2 EFB	3 EC	3 ECD		4 EF
F	4 FEA	4 FB	4 FEC	4 FD	4 FE	

(b)

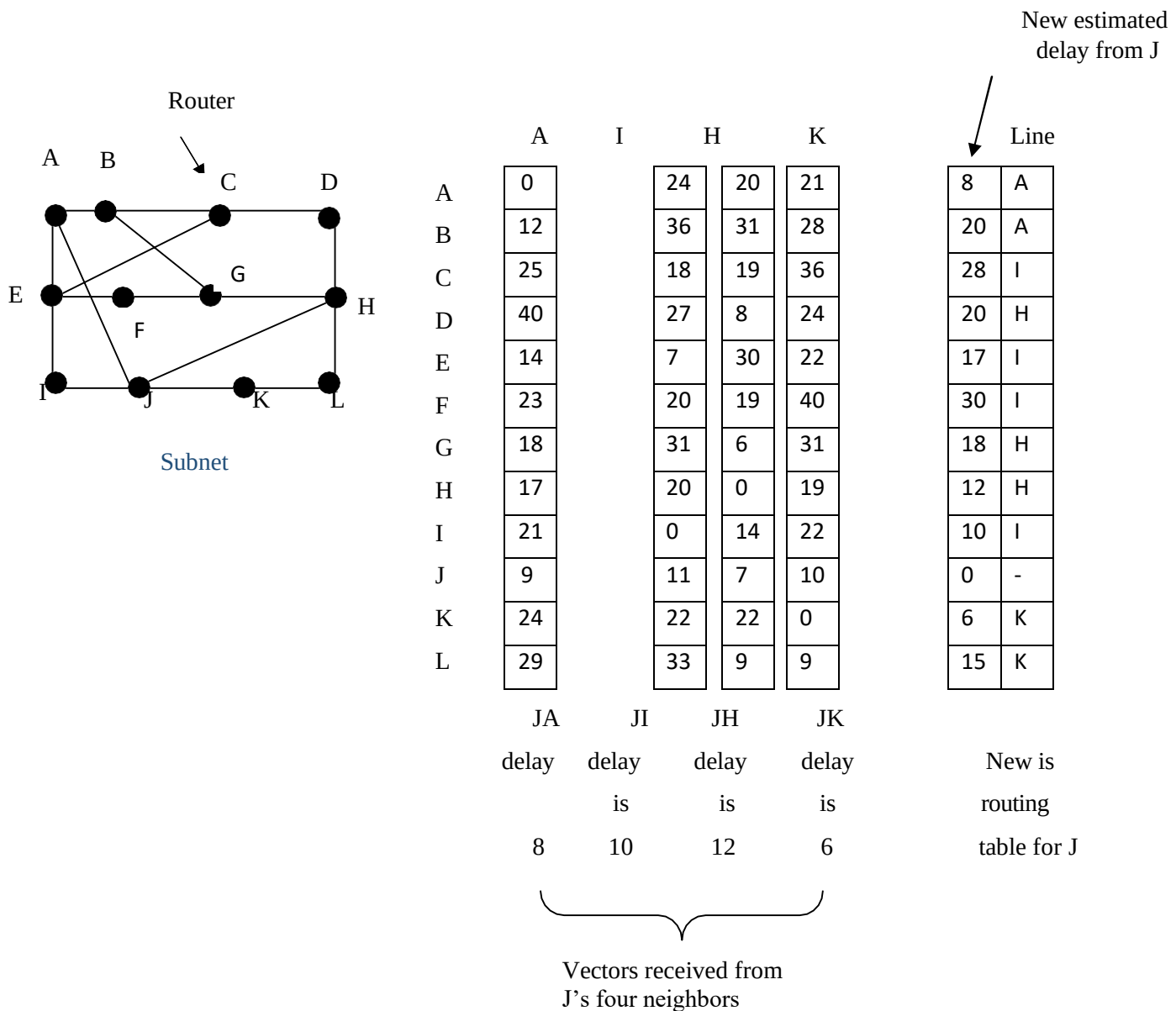
Distance Vector Routing:

This is a dynamic routing algorithm. This algorithm operates by having each router maintain a table (i.e. a vector) giving the best known distance to each destination and which line to use. These tables are updated by exchanging information with the neighbors.

The routing table indexed by and containing one entry for each router in the subnet. This entry contains two parts: The preferred outgoing line to use for the destination and an estimate of time or distance to that destination. The metric used might be number of hops, time delay in msec, total number of packets queued along the path or something similar.

The router is assumed to know the distance to each of its neighbors. If the metric is hops, the distance is just one hop. If the metric is queue length, the router examines each queue. If the metric is delay the router can measure it directly with a special ECHO packets.

Consider an example, in which the delay is used as metric and the router knows the delay to each of its neighbors. Once every T msec each router send to each neighbor a list of its estimated delays to each destination. It also receives a similar list from each neighbor. Let x_i being x 's estimate of how long it takes to get router 'i'. If the router knows that the delay to x is 'm' msec. To get router i via x is $(x_i + m)$ msec. By performing this calculation for each neighbor, a router can find out which estimate is the best and use that estimate and the corresponding line in its new routing table.



Input from A,I,H, K and new routing table for J

Fig.(a) shows the subnet and fig.(b) shows the vectors of J for its neighbors. Fig.(c) shows the new routing table for J. Let JA delay is 8, JI delay is 10, JH is 12, JK is 6.

The new route to G from J can be calculated as follows. J can get A in 8 m sec.

A can get G in 18 m sec(from table)

□ J can get G in $(8+18)$ 26 m sec.

Similarly the delay to G via I,H and K is $(31 +10)$ 41, $(6+12)$ 18, $(31+6)$ 37 m sec.

The best of these values is 18, so it makes an entry in its routing table that the delay to G is 18 m sec and that route is via H.

Hierarchical Routing:

As network grow in size, the router routing tables grow proportionally. Not only more memory consumed by ever increasing tables, but more CPU time is needed to scan them more bandwidth is needed to send status reports about them. At a certain point the network may grow to the point where it is no longer feasible for every router to have an entry for every other router, so the routing will have to be done hierarchically.

When hierarchical routing is used, the routers are divided into ‘Regions’. Each router knows all the details about how to route packets to destinations within its own region, but doesn’t know the internal structure of other regions. For huge networks, a two-level hierarchy may be insufficient, it may be necessary to divide the regions into clusters, clusters into zones and zones into groups and soon.

Consider a two level hierarchy with five regions as shown in fig. one router needs 17 entries for one table. The network contains 17 routers. So the total no. of entries will be 17 x 17. This is for when we are not using hierarchy.

When routing is done hierarchically a router will consists of entries for all the local routers and regions only.

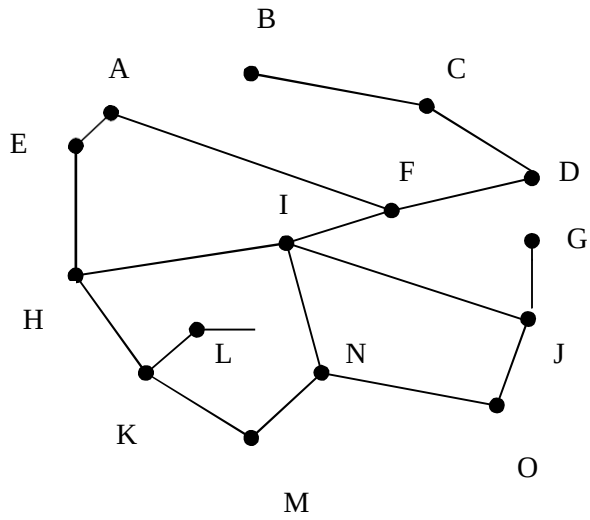
For Ex. The router 1A consists of entries as shown in fig.(c). Hierarchical routing has reduced the table from 17 to 7 entries.

Full table for 1A			Hierarchical table for 1A		
Region 1		Region 2	Dest.	Line	Hops
1A	1B		-	-	
	1C		1B	1	
1A	1C		1C	1	
			1B	2	
			1C	2	
			1C	3	
			1C	4	

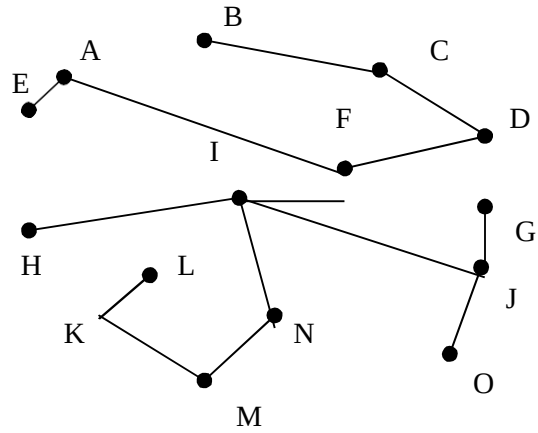
4) Another algorithm which uses the spanning tree. A spanning tree is a subnet of the subnet that includes all the routers but contains no loop. If each router knows which of its lines belong to the spanning tree, it can copy an incoming broadcast packet onto all spanning tree lines except the one it arrived on. This method makes use of bandwidth excellently and generates minimum no. of packets necessary to do the job. The only disadvantage is that each router must have knowledge of some spanning tree.

5) One more algorithm is an attempt to approximate the behavior of the previous one, even when the routers do not know anything at all about spanning trees. The idea is remarkably simple once it has been pointed out. When a broadcast packet arrives at a router, the router checks to see if the packet arrived on that line that is normally used for sending packets to the source of the broadcast. If so, there is an excellent chance that the broadcast packet itself followed the best

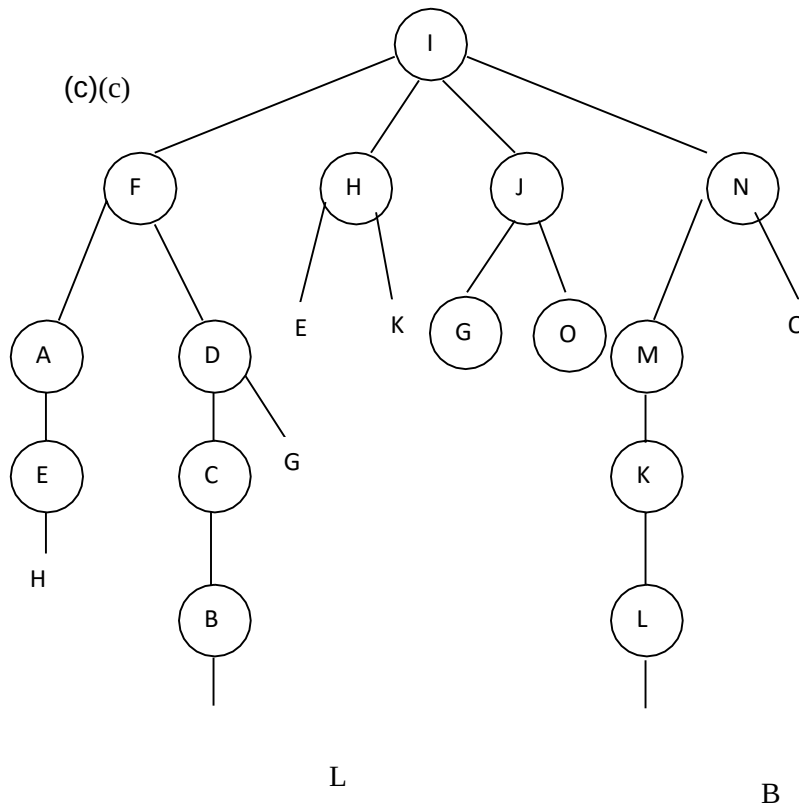
route from the router and is therefore the first copy to arrive at the router. This being the case, the router forwards copies of it onto all lines except the one it arrived on. If, however, the broadcast packet arrived on a line other than the preferred one for reaching the source the packet is discarded as a likely duplicate.



(a)



(b)(b)



Multicast Routing :

For some applications, it is necessary for one process to send a message to all other members of the group. If the group is small, it can just send each other member a point-to-point message. If the group is large this strategy is expensive. Some times broad casting is used, but using broad casting is used, but using broadcasting to inform 1000 machines on a million node network is inefficient because most receivers are not interested in the message. Thus it is needed to send message to well-defined groups. Sending message to such a group is called ‘multicasting’.

To do multicasting, group management is required. Some way is needed to create and destroy groups and for processes to join and leave groups. When process joins a group, it informs its host of this fact. It is important that routers know which of their hosts belong to which group. Either hosts most inform their routers about change in group membership or routers must query their hosts periodically. Routers tell their neighbors, so the information propagates through the subnet.

To do multicast routing, each router computes a spanning tree covering all other

routers in the subnet. When a process sends a multicast packet, to a group, the first router examines its spanning tree and prunes it, removing all lines that do not lead to hosts that are members in the group. Multicast packets are forwarded.

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